I have been targeting applied AI within healthcare roles, and it's rare to find one deployed to this degree. I wouldn't want to work in theoretical AI research without any actual development and evaluation in the real world. This role is very clearly in that space. The size of the group, along with its diverse expertise, makes me confident that I would be able to contribute fast and learn even faster. Whether it's figuring out the clinical side or even implementation feasibility in a hospital, I know I'll have the resources to learn from when needed, and the rest is up to me.

I have been moving in this direction during my master's, especially through work on healthcare data systems and the gap between technical capability and clinical reality. Taking courses like healthcare innovation to learn and learning about MDR & how the AI Act affects healthtech innovation in the EU. I've always leaned towards R&D in my work instead of the operations side of a product, and growing up in a compound for doctors, I picked up both an interest and a good idea of certain systems.

A lot of my experience has involved working with messy, real-world data and figuring out what needs to change before modelling even makes sense. In an ICU collaboration with a hospital in Saudi Arabia, I worked on a project to build internal dashboards for KPI tracking. Eventually, it turned into redesigning how clinical data was collected, due to faults in the original form design, re-training of nurses and defining keywords/phrases that differ from textbook to a specific department's daily language. All these things were discovered and had to be fixed for a useful dashboard to be built.

It's a given that the majority of active time spent towards an AI model deployment isn't in picking models or optimisation, which still has its place. But figuring out the data origins and ETL construction, learning about the domain and use-cases. You make so many decisions and assumptions, both consciously and unconsciously, when designing an AI system. It's why I've started pushing for regular short stakeholder meetings in my more recent projects, even if they haven't mentioned it. You need active and interested parties to make the best product. It's because of these experiences that I think I would do well in your group, especially with the first responsibility described.

More recently, I worked on a consulting-style project with a healthtech company working on EHR systems at Akademiska Sjukhuset in Uppsala. The mess that is interoperability in healthcare and how that issue is present across Sweden. The various standards are utilised to varying degrees across regions, hospitals and even departments within the same hospital. I conducted interviews with clinicians and IT staff about interoperability issues, with the consistent theme being that technical solutions often fail at the integration stage. This experience should make it easier for me to work on the system integration aspects.

I've been interested in fair and responsible AI for a while. You can see my old Medium article on how the proven biases in healthcare research due to predominantly white male test subjects would be propagated further by AI models being trained on this same ideology. It's surprising how long it took for sex-stratified trial results and other basics for health equity in research to be required.

On the technical aspects, I'm very comfortable with Python, statistics and analytics. I do lack experience in any deployment via containers or Kubernetes, aside from the basics I learned in my courses. Similarly, I haven't worked in a real-world image analysis project for some time. However, just last week, I began a small pilot research project at King Faisal Hospital in Saudi Arabia, working on using AI to read, suggest and follow through with electrophoresis readings. Specifically, to reduce the turnaround time for lab results for simpler ones, like the M-test for myeloma.

I would be quite interested in the clinical evaluation side of the role, too. Every new domain I work in has its own set of targets and tests used to give them confidence in deployment. There are a lot of metrics and tests you can utilise for post-deployment analytics, and learning how they interact and why one works best in practice is one of the things that makes you a good AI engineer.

I am also interested in the longer-term possibility of continuing to a doctoral position, but I am applying primarily because the work itself makes sense to me. If the system you are building is meant to support real clinical decision-making, then the difficult part is not only the model but how it fits into practice. That is the part I want to work on.